Fundamental theorem two

If $f(x)$ is continuous, and $G(x) = \int_a^x f(t)\,dt \quad (a \le t \le x)$

then $G'(x) = f(x)$

$G(x)$ solves the differential equation

$$\begin{cases} y' = f \\ y(a) = 0 \end{cases}$$

$G(x) = F(x) + C$

$G(a) = \int_a^a f(t)\,dt = 0$

$G(a) = F(a) + C = 0$

$C = -F(a)$

$\int_a^b f(t)\,dt$

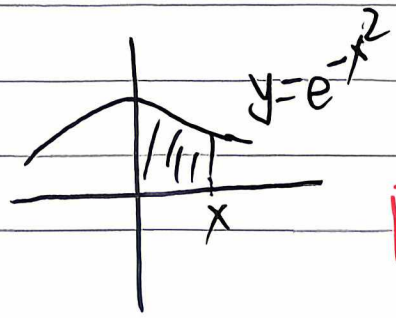
$G(x) = F(x) - F(a)$

$x = b \quad G(b) = F(b) - F(a) = \cancel{\int_a^b f(t)\,dt}$

$$\frac{d}{dx}\boxed{\int_1^x \frac{dt}{t^2}} = \frac{1}{x^2} \qquad G'(x) = f(x) = \frac{1}{x^2}$$

$\downarrow$

$G(x) \qquad \int_1^x f(t)\,dt, \quad f(t) = \frac{1}{t^2}$



$F'(x) = y \quad F(x) = \int_0^x e^{-t^2}\,dt$

$F(x)$ can't be expressed in terms of logs, exp, trig ---

Claim $L(ab) = L(a) + L(b)$ $\quad L(x) = \int_1^x \frac{dt}{t}$

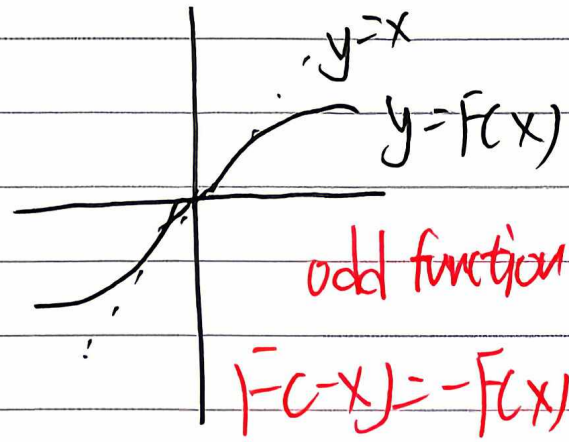
$$L(ab) = \int_1^{ab} \frac{dt}{t} = \int_1^{a} \frac{dt}{t} + \int_a^{ab} \frac{dt}{t}.$$

$$\int_{t_1=a}^{t_2=ab} \frac{dt}{t} = \int_{u_1=1}^{u_2=b} \frac{a\,du}{au} = L(b)$$

$t = au \qquad t_1 = au_1 = a \Rightarrow u_1 = 1$

$dt = a\,du \qquad t_2 = au_2 = ab \Rightarrow u_2 = b$

$\star\ F(x) = \int_0^x e^{-t^2} dt$



odd function

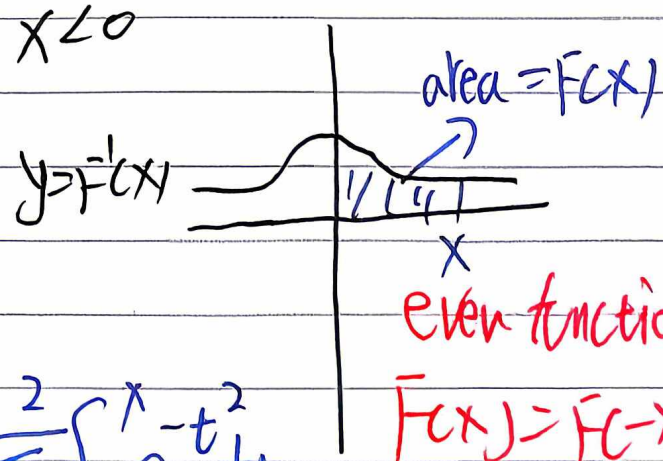
$F(-x) = -F(x)$

$F'(x) = e^{-x^2} \quad F'(0) = 1$

$F''(x) = -2x \cdot e^{-x^2} \quad \begin{cases} <0 & x>0 \\ >0 & x<0 \end{cases}$

$\lim_{x \to +\infty} F(x) = \frac{\sqrt{\pi}}{2}$

$x \to -\infty \quad F(x) = -\frac{\sqrt{\pi}}{2}$



even function

$F(x) = F(-x)$

Error function $\quad \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

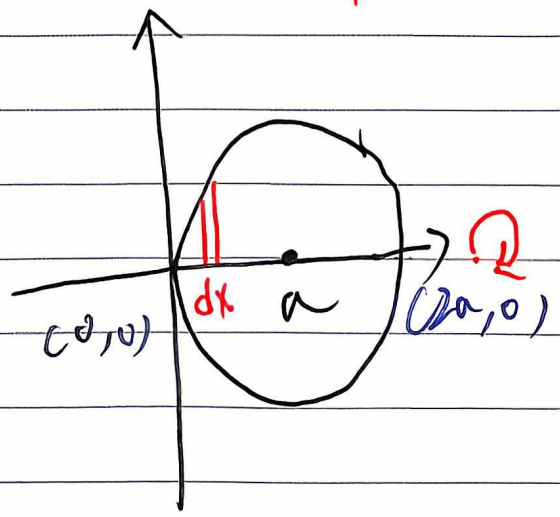
Volumes by slicing

$\Delta V \approx A \Delta x$

$dV = A(x)\,dx$ one slice

$V = \int A(x)\,dx \approx \sum A_i \Delta x$

① method of disk



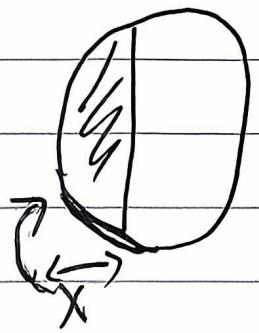
$(x-a)^2 + y^2 = a^2 \quad y^2 = 2ax - x^2$

$dV = \pi r^2 dx = \pi y^2 dx$

$V = \int \pi (2ax - x^2)\,dx$

revolve around x-axis $\quad V = \pi \left(ax^2 - \frac{x^3}{3}\right)\Big|_0^{2a}$

$0 < x < 2a \qquad = \frac{4}{3}\pi a^3$



$V(x) = \pi \left(ax^2 - \frac{x^3}{3}\right)$

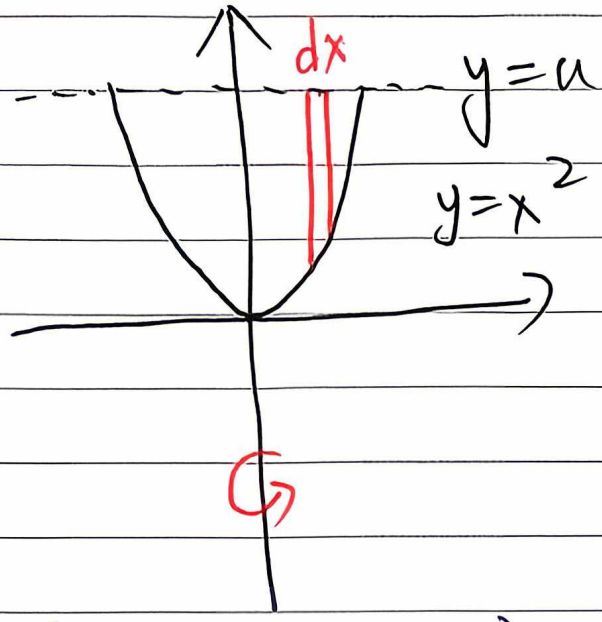
$V(a) = \pi \left(a \cdot a^2 - \frac{a^3}{3}\right)$

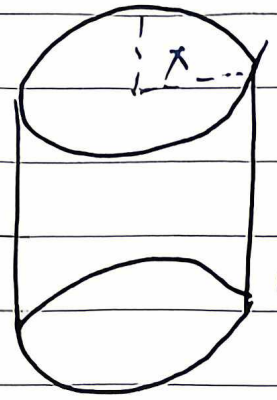
$= \frac{2}{3}\pi a^3$

half

$dx \perp$ x-axis $\Longrightarrow$ disk

② method of shells





thickness = dx

height = a − y

(y_top − y_bottom)

$$dV = (2\pi x)(a-y)\,dx$$
$$= 2\pi x(a-x^2)\,dx$$
$$= 2\pi(ax-x^3)\,dx$$

relove around y-axis

$0 < x < \sqrt{a}$

$$V = \int_0^{\sqrt{a}} 2\pi(ax-x^3)\,dx = 2\pi\left(\frac{ax^2}{2}-\frac{x^3}{4}\right)\Big|_0^{\sqrt{a}}$$
$$= 2\pi\left(\frac{a^2}{2}-\frac{a^2}{4}\right) = \frac{\pi}{2}a^2$$

dx // y-axis ⟹ shell